ORCHESTRATION, INTEGRATIONS, SECURITY, COST

Scalability & Production Readiness

Janos Kristof · July 15, 2026

Prepared under NDA · call data anonymized

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Rep Coach — Production Readiness

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Rep Coach — Production Readiness

Supporting deep-dive · Janos Kristof · *what it takes to run the prototype for real — safely, and without me*

> The prototype proves the loop works. This is the honest answer to the three questions a VP of Ops asks next: **how does it trigger itself, how does confidential call data stay safe, and how does it keep running long after I've moved on?** Scope and design are in `phase2_build/`; the component-level buy-vs-build framework is in `phase1_diagnose/make_vs_buy.md`.

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1. From prototype to production — what's mocked, what's real

Everything load-bearing in the scoring loop is already real; what's mocked is the **plumbing around it**, and each mock has a clean production swap because it sits behind an interface.

Concern	Prototype (today)	Production
Trigger	Manual **Process** button	Jiminny **call-complete webhook** (§4)
Scoring core	Real — `scoring.py` (deterministic + Claude)	**Unchanged** — the differentiator ships as-is
Routing	`MockNotifier` / `MockCRM` → local JSON log	`SlackNotifier` / `HubSpotCRM` behind the same `Notifier` / `CRM` seam (§3)
Surface	Streamlit app	**HubSpot** — scorecard on the call record + tasks + dashboards (§3)
LLM endpoint	Anthropic API, synthetic text only	Zero-retention **BAA** endpoint, real (PII-bearing) transcripts (§5)
Store	`data/*.json`	CRM record + a small evaluations store; versioned rubric
Access	Single local user	**Auth / RBAC** — rep sees own, manager sees team (§5)

The design intent from day one (`design.md` §6) was that **capture is bought, the loop is built, and routing is an adapter swap** — so "productionizing" is wiring, not a rewrite.

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2. Orchestration decision — decided, with reasoning

The question: once Jiminny fires "call complete," what runs the sequence *get transcript* → *score* → *branch* → *route* → *persist*? This was the one row left **OPEN** in `make_vs_buy.md` (now closed to the decision below). I evaluated it against the make-vs-buy criteria that matter for a **one-person ops function at a HIPAA-adjacent, Series-C company**: NDA/data-residency fit, testability + prompt/version control, observability, maintenance-by-one-person, and lock-in.

Option	Data-residency / NDA	Testable LLM logic	One-person ops	Verdict
No-new-tool — HubSpot Workflows / Operations Hub + a small service	Good (HubSpot already holds the data)	Weak in-platform; core stays in the service	**Best** — no new surface	**Test first**
n8n (self-hosted)	**Excellent** — self-hostable, in-region	Weak in-node; fine if it *calls* the core	Another runtime to run/monitor	**Fallback when complexity grows**
Zapier / Make	Weaker — SaaS data path under NDA/HIPAA	Weak	Low, but cost + residency concerns at scale	Rejected
Temporal / Windmill / Prefect	Good (self-host)	**Strong** (code-first)	Heaviest to operate	Overkill now
Custom Python service	Full control	**Strong**	Most to maintain	The core *is* this already

Decision. Keep the scoring core in code — it must stay testable (the golden-transcript check), prompt-versioned, and robust to structured-output quirks (all hardened in Phase 2). Then **use the lightest orchestration that clears NDA/data-residency**: since **HubSpot is confirmed as the CRM**, test **HubSpot Workflows / Operations Hub + a thin scoring service** *first* — it adds no new vendor and keeps the data in one place.

When a platform earns its keep. An orchestration platform (self-hosted **n8n** is the strongest NDA-friendly candidate) is justified **when complexity grows** — many triggers/signal sources, many teams × rubrics running in parallel, branching human-in-loop approvals, and retries/monitoring at scale. That's a *reach-for-it-when-justified* option, not a now-decision. Critically, adopting it later is a **glue change, not a rewrite**: the `Notifier` / `CRM` adapter seam means the Python core is reused unchanged and n8n's Slack/HTTP nodes simply implement the same interface. *We are not locked out of it, and we are not locked into it.*

> **Anti-pattern rejected:** re-implementing scoring inside workflow Function nodes (deterministic math in JS + a raw LLM call). It throws away the tests and the structured-output hardening — the exact

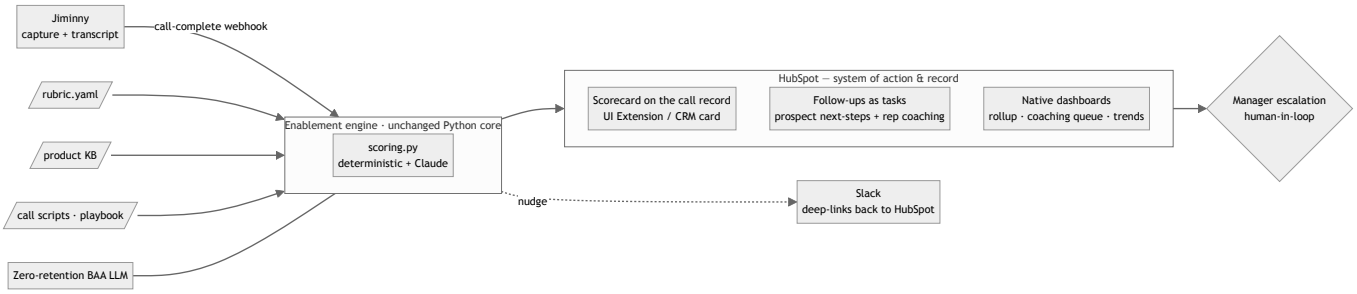
> things that make the scores trustworthy. Buy the commodity, build the differentiator, keep the differentiator in code.

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3. Production architecture — HubSpot as the single surface

The strongest fix for the multi-tool jump (**pain #1**) is to stop adding surfaces and make **HubSpot the one place the work lives.** The scoring engine is unchanged; only the adapters and the trigger become real.

> **The concrete HubSpot mechanics** — how each Rep Coach output maps to a HubSpot object (custom object · CRM card · Tasks · workflows · native dashboards), the prototype→production adapter mapping, the tier caveats, and the discovery questions — are worked in the companion deep-dive `hubspot_surface.md`.



- **Scorecard on the call record** — the Jiminny call is an engagement on the contact/deal; the evaluation is written back via a **HubSpot UI Extension / CRM card** (or a linked custom object). Visibility follows RBAC: rep sees own, manager sees team.
- **Follow-ups as HubSpot tasks — two kinds:** prospect next-steps (dated tasks on the deal) *and* rep self-improvement to-dos (coaching, linked to micro-content). Routed by HubSpot Workflows.
- **Dashboards in HubSpot** — team rollup, coaching queue, per-rep/-dimension trends as native reports (the make-vs-buy "BI → buy later" verdict cashes out here; Streamlit was the prototype surface only).
- **Slack = nudge layer** — "your call was scored," "3 reps need coaching" — deep-linking back into HubSpot, not a second place to do the work.
- **Grounding artifacts feed the loop** — the **product KB** grounds proof-fidelity scoring and supplies recommended micro-content; **call scripts/playbook** ground motion-adherence scoring.

The prototype's `Notifier` / `CRM` protocols (`design.md` §6) are exactly the seam: `SlackNotifier` and `HubSpotCRM` replace the mocks; **the core never changes.**

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4. The trigger — no manual upload

Replace the manual Process button with a **Jiminny call-complete webhook**: Jiminny fires when a call is transcribed → the orchestration layer fetches the transcript via Jiminny's API → the scoring service runs → results write back to HubSpot and route the coaching loop. The rep and manager see the scorecard **on the record** minutes after the call ends, with zero manual steps. (Discovery item: confirm Jiminny's webhook + transcript-API availability on Sensi's plan — §10.)

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5. Data safety & security — the NDA + HIPAA posture

Sensi handles senior-care data; call transcripts are confidential and HIPAA-adjacent. This is the non-negotiable layer.

- **LLM endpoint** — a **zero-retention** contract with a **BAA** in place: Anthropic (zero-retention
- BAA), **AWS Bedrock**, or **Azure OpenAI** (both offer BAA + in-VPC / in-region). Never build a model; the only thing that matters is the terms. *The prototype sends only synthetic text — no real PII has left the machine.*
- **PII handling** — transcripts stay in-region; optional PII redaction before the LLM call for fields not needed for scoring; encryption in transit + at rest.
- **Auth / RBAC** — rep sees own scorecards, manager sees team, high-stakes coaching scoped appropriately; inherited from HubSpot's permission model where the surface is HubSpot.
- **Audit** — every evaluation carries `rubric_version` + `model` (already in the record), so any score is reproducible and attributable — the basis of an audit trail.

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6. Human-in-loop — the automated/human boundary

The loop **automates the mechanical** (scoring, routing, surfacing) and **keeps judgment human**:

- **Automated:** score every call, attach the scorecard, route nudges + draft coaching tasks, compute trends.
- **Human:** manager **escalations** (a very low score or a fired flag surfaces a call, a person decides), **rubric changes** (what "good" means is a human call, versioned), and **high-stakes coaching** conversations. The tool triages and drafts; it never delivers the hard conversation.

This is the same boundary Phase 3 carries across teams.

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7. Production make-vs-buy — finalized

`make_vs_buy.md`'s leans hold; this session **closes the one OPEN row**:

- **Orchestration** — was OPEN → **decided (§2):** scoring core in code; HubSpot-native orchestration first, self-hosted n8n as the complexity-driven fallback.
- **Everything else stands:** capture = **buy (Jiminny)**; scoring loop = **build (the differentiator)**; KB/scripts, training, BI = **buy/config**; surface = **consolidate onto HubSpot**; LLM = **buy with BAA terms**.
- **Tier-dependent caveat to confirm in discovery:** HubSpot **UI Extensions / custom objects / Operations Hub** depend on the licensed tier (Enterprise / developer projects). If a capability isn't licensed, the fallback is a linked custom object + a lightweight embedded view — the single-surface principle holds, the mechanism changes.

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8. Cost model — order-of-magnitude, not false precision

- **LLM** — one structured call per scored call. At Claude Sonnet-class pricing and a typical transcript, cost is **cents per call** (~$0.02–0.10). At, say, 2,000 calls/month that's roughly **$40–200/month** — negligible against a single rep's ramp. Deterministic metrics are free (pure code).
- **Orchestration / hosting** — HubSpot-native adds no new license beyond the existing tier; a small scoring service is a minimal container (~tens of $/month). Self-hosted n8n, if adopted later, is a single small VM.
- **HubSpot tier** — the only potentially material line: UI Extensions / Operations Hub may require a tier step-up — priced during discovery.
- **Monitoring** — commodity (logs + alerts), negligible.

The dominant cost is **not** the software — it's keeping the rubric and KB current, which is human time, and the entire point of the tool is to *reduce* the larger cost (manager review hours + slow ramp).

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9. Monitoring & "runs without me"

> **The LLM trust mechanics** — the deterministic anchor, forced structured output, quote verification, prompt-injection stance, and versioned-rubric change control — are worked in the companion deep-dive `guardrails.md` (deck S16 "Guardrails" + the S19 appendix).

The assignment's soft ask — *keeps running long after I've moved on* — is an operational design goal:

- **Versioned rubric** — every eval stamps `rubric_version`; rubric changes are reviewable, revertible.
- **Structured-output health** — monitor LLM schema-adherence + the non-verbatim-quote soft check (`verify_quotes`) already in the core; a malformed-output rate crossing a threshold alerts. An **error queue** holds calls that failed to score for retry, so nothing is silently dropped.
- **Config over code** — dimensions, weights, motion, proof wall, role, and flags all live in YAML; a new team is a new rubric (proven by AC8), so the common change needs no engineer.
- **One-command deploy + one-command re-score** — reproducible, documented; the same discipline the prototype already follows (fixed seed, baked dates).
- **Human owner, not a hero** — because the core is small, testable, and config-driven, a single ops person maintains it; there is no bespoke infra to babysit.

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10. Open questions — resolved in discovery

- **HubSpot tier / licensing** — are UI Extensions / custom objects / Operations Hub available? → shapes the write-back surface (§3, §7).
- **Jiminny webhook + transcript API** — available on Sensi's plan? → decides how automated the trigger is (§4) and how much is *configure* vs *build* (`make_vs_buy.md`).
- **BAA status** — which LLM vendor already has (or will sign) a BAA with the right zero-retention / data-residency terms? → gates the production go-live (§5).

> **The through-line:** productionizing Rep Coach is **integration and governance, not reconstruction.** The differentiator already runs and is tested; production is wiring the real trigger, swapping the mock adapters for HubSpot/Slack behind the seam that was built for it, and putting the NDA/HIPAA and operational guardrails in place — the work a real enablement lead does to make one good loop safe, single-surfaced, and durable.